SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME:CSA1610 **Student Name:-D. Deepak**
COURSE CODE: Data Warehousing and Data Mining **REG NO:-192572095**

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

QUESTIONS: 05 **Total Marks:50**

Annexure A

Data Interpretation

Criteria	Understanding of Data ***(2.5)***	Analysis ***(2.5)***	Application of Concepts ***(2)***	Conclusion ***(1.5)***	Clarity ***(1)***	***Total(10)***
Weightage	*25%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I D.Deepak/192572095 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: deepak

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values:-

K	SSE
1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):
A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

ANS:-

Hospital Patient Segmentation — K-Means

The healthcare organization uses **K-Means clustering** to group patients according to their medical expenses and health-risk scores. The **Elbow Method** is used to identify the suitable number of clusters.

K	SSE	Reduction in SSE
1	600	—
2	350	250
3	220	130
4	200	20
5	190	10

SSE represents the total squared distance of data points from their respective cluster centers. A lower SSE means that the data points are closer to their cluster centers.

From the given values, SSE decreases rapidly from **600 at K = 1 to 220 at K = 3**. After K = 3, the reduction becomes very small. The reduction from K = 3 to K = 4 is only 20, and from K = 4 to K = 5 is only 10.

Therefore, the elbow point occurs at K = 3, and the optimal number of clusters is 3.

Patient Segmentation:-

The three clusters can broadly represent:

- **Cluster 1 – Lower-risk patients:** Patients requiring comparatively less medical attention.
- **Cluster 2 – Moderate-risk patients:** Patients who may require regular monitoring and preventive care.
- **Cluster 3 – Higher-risk patients:** Patients who may require closer monitoring and more healthcare resources.

For example, P1 and P4 have relatively lower medical expenses and high risk scores, whereas P2 and P5 have higher expenses but lower risk scores. This shows why clustering considers multiple attributes rather than only one factor.

Effect on Healthcare Decision-Making:-

Selecting three clusters helps the organization avoid excessive fragmentation of patients. If **K = 1**, all patients would be treated as one group, hiding important differences. If too many clusters are selected, the groups may become unnecessarily small and difficult to manage.

The clusters can support:

1. Patient risk identification.
2. Personalized healthcare planning.
3. Preventive-care programs.
4. Allocation of doctors and hospital resources.
5. Medical budget planning.
6. Identifying patients requiring regular monitoring.
7. Designing suitable healthcare services.

Conclusion: The optimal number of clusters is **K = 3** because the SSE decreases significantly up to K = 3 and shows only marginal improvement afterward. This provides meaningful patient segmentation while keeping the clustering model simple and useful for healthcare decision-making.

Q2. Dataset: E-Commerce Order Analysis

Order ID	Product Category	Order Value (₹)
O1	Electronics	55000
O2	Clothing	25000

Order ID	Product Category	Order Value (₹)
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):
An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

ANS:- **E-Commerce Order Analysis — Hierarchical Clustering**

Hierarchical clustering is an unsupervised learning technique that creates a hierarchy of similar data points. The resulting hierarchy is represented using a **dendrogram**.

The dataset contains product category and order value:

Order	Category	Order Value
O1	Electronics	₹55,000
O2	Clothing	₹25,000
O3	Electronics	₹57,000
O4	Furniture	₹30,000
O5	Clothing	₹27,000

In hierarchical clustering, similar observations are first joined together. As the clustering process continues, smaller groups are merged into larger groups.

Interpretation of the Dendrogram

O1 and O3 are highly similar because:

- Both belong to Electronics.
- Their order values are very close.
- Their values are also relatively high.

Therefore, **O1 and O3 form one cluster**.

Similarly, O2 and O5 are both Clothing orders and have similar values. Therefore, they form another cluster.

O4 belongs to Furniture and has a different category and purchasing value, so it can be treated as a separate cluster.

Thus, an appropriate three-cluster interpretation is:

- **Cluster 1:** O1, O3 → High-value Electronics

- **Cluster 2:** O2, O5 → Clothing
- **Cluster 3:** O4 → Furniture

Therefore, **3 clusters** provide a reasonable interpretation of the given dataset.

Significance of the Clusters:-

These clusters help the retailer understand different purchasing patterns.

The Electronics cluster indicates customers or orders involving **high-value products**. The Clothing cluster represents comparatively lower-value purchases with similar behavior. Furniture represents a separate purchasing category.

The company can use this information for:

1. Targeted advertisements.
2. Product recommendations.
3. Category-specific discounts.
4. Inventory management.
5. Sales forecasting.
6. Customer segmentation.
7. Identifying high-value purchases.
8. Designing promotional campaigns.

A dendrogram also provides flexibility. If the company wants broad segmentation, it can select fewer clusters. If more detailed segmentation is required, the dendrogram can be cut at a lower level.

Conclusion: Hierarchical clustering provides a clear visual representation of similarities between orders. Three clusters provide meaningful groups and help the retailer understand product preferences and purchasing behavior.

Q3. Dataset: Mobile App User Behavior

User	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):
A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based

membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

ANS:- Mobile App User Behavior — Soft Clustering

The mobile application company wants to segment users based on daily usage time, login frequency, and in-app purchase amount.

The given data is:

User	Usage Time	Login Frequency	Purchase
U1	120 min	18	₹2,000
U2	60 min	5	₹5,000
U3	140 min	22	₹1,500
U4	50 min	4	₹6,000
U5	130 min	20	₹1,800

The users clearly show different behavioral patterns.

- U1, U3 and U5 have high usage time and frequent logins.
- U2 and U4 have lower usage and login frequency.
- U2 and U4, however, have relatively high in-app purchases.

This means that a user may not belong completely to only one behavioral group.

Probability-Based Membership

Soft clustering assigns each user a membership probability for different clusters.

For example:

User	Active User	Occasional User	High-Spending User
U1	0.80	0.15	0.05
U2	0.20	0.25	0.55
U3	0.90	0.08	0.02
U4	0.10	0.20	0.70
U5	0.85	0.12	0.03

These values are illustrative; actual probabilities would be calculated by the selected soft-clustering algorithm.

U3 has a very high membership in the active-user group. U4 has a high membership in the high-spending group. U2 is less active but contributes significant revenue through purchases.

Soft vs Hard Clustering:-

In hard clustering, a user belongs to only one cluster. In soft clustering, a user can have partial membership in several clusters.

Soft clustering is useful because customer behavior is often overlapping. A user can be both an active user and a high-value customer to some degree.

Advantages:-

1. Represents overlapping behavior.
2. Handles uncertainty.
3. Provides more detailed customer profiles.
4. Helps identify valuable but less-active users.
5. Supports personalized recommendations.
6. Helps design targeted marketing campaigns.
7. Improves customer retention strategies.
8. Helps identify different revenue patterns.

For example, U4 may not be an active user, but the high purchase amount makes this user commercially important.

Conclusion: Soft clustering provides a more realistic representation of customer behavior because users can have different degrees of membership in multiple groups. It is therefore useful for personalized marketing and customer behavior analysis.

Q4. Dataset: Agricultural Crop Monitoring

Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):
An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

ANS:- Agricultural Crop Monitoring — Normalization

The agricultural research center performs clustering using rainfall, soil fertility, and crop yield.

Farm	Rainfall	Soil Fertility	Crop Yield
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

The three variables have significantly different scales. Rainfall ranges from 280 to 520, soil fertility ranges from 3 to 9, and crop yield ranges from 65 to 90.

Clustering Before Normalization:-

Without normalization, distance-based clustering can give excessive importance to variables with larger numerical ranges.

For example, the numerical difference in rainfall between two farms can be much larger than the difference in soil fertility. As a result, rainfall may dominate the clustering process.

This can lead to clusters that are based mainly on rainfall rather than the overall agricultural characteristics.

Clustering After Normalization:-

Normalization converts the variables into a comparable scale. Therefore, rainfall, soil fertility, and crop yield can contribute more equally to the clustering process.

The farms can be interpreted as:

- F1 and F4: Higher soil fertility and higher crop yield.
- F2 and F5: Higher rainfall but lower soil fertility and lower yield.
- F3: Intermediate agricultural characteristics.

Importance of Normalization

Normalization improves the reliability of clustering because it prevents one feature from dominating due to its numerical scale.

It provides:

1. Fair contribution from all variables.
2. Better distance calculations.
3. More meaningful clusters.
4. Reduced scale-related bias.
5. Improved interpretation of agricultural patterns.
6. Better comparison between farms.

7. More reliable crop analysis.
8. Better decision-making.

For example, after normalization, the research center can identify farms that have similar overall agricultural conditions, rather than simply grouping farms according to rainfall.

The results can support decisions related to irrigation, fertilizer usage, crop selection, resource allocation, and yield improvement.

Conclusion: Normalization is an important preprocessing step for clustering agricultural data. It makes the features comparable and produces more reliable and meaningful farm clusters.

Q5. Dataset: Employee Performance Evaluation

Employee	Productivity Score	Attendance (%)	Skill Assessment Score
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model	Score
K=2	0.48
K=3	0.68
K=4	0.52

Question (BL5 – 8 Marks):
A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

ANS:-Employee Performance Evaluation — Silhouette Score

The company uses clustering to group employees based on productivity, attendance, and skill assessment scores.

The silhouette scores are:

Number of Clusters	Silhouette Score
K = 2	0.48
K = 3	**0.68**
K = 4	0.52

The Silhouette Score evaluates how well each employee fits within their assigned cluster compared with other clusters.

A higher score indicates that the clusters are generally more compact internally and better separated from one another.

Analysis:-

For K = 2, the score is **0.48**, which indicates reasonable but less effective separation.

For K = 3, the score increases to **0.68**, which is the highest among the three models.

For K = 4, the score decreases to **0.52**, showing that adding another cluster does not improve the grouping.

Therefore:

Optimal number of clusters = K = 3

Employee Segmentation

The three clusters can be interpreted as:

Cluster 1 – High Performance:
Employees with high productivity, attendance, and skill scores.

Cluster 2 – Moderate Performance:
Employees with average performance and skill levels.

Cluster 3 – Lower Performance:
Employees who may have comparatively lower productivity, attendance, or skill scores.

For example:

- **E5:** Productivity 93, Attendance 96%, Skill 90 → likely high-performance group.
- **E4:** Productivity 55, Attendance 65%, Skill 58 → likely lower-performance group.
- **E3:** Productivity 78, Attendance 88%, Skill 72 → likely moderate-performance group.

Importance to the Company

The selected clustering model can help management:

1. Identify high-performing employees.
2. Find employees requiring training.

3. Plan employee development programs.
4. Provide suitable rewards and recognition.
5. Identify performance gaps.
6. Improve workforce planning.
7. Support data-driven HR decisions.
8. Monitor employee performance over time.

The K = 3 model is preferable because it provides the **highest silhouette score of 0.68**, indicating better-defined clusters than K = 2 or K = 4.

Conclusion: The most suitable model is **K = 3**. Its highest silhouette score shows that three employee groups provide the best separation and meaningful performance segmentation for the organization.